

Analysis of On-Time Performance of the US Aviation Market

DATA 601 – L02 – T7

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1. Introduction

Frequent flight delays and cancellations are a major challenge in the U.S. aviation industry, disrupting travel plans, causing financial losses, and reducing passenger satisfaction. These issues impact not just travel, airports, and the economy, highlighting the need for effective solutions. Given that the US aviation market is ranked first globally as "Atlanta's Hartsfield-Jackson International Airport ranked as the busiest airport in the world consistently for two decades, with 104.7 million passengers passing through in 2023" it has significant impact on our day to day lives. During the planning stage of this project, we formulated key guiding questions that we hoped could explain the causes of delay, how on-time performance differs across different airlines, airports, and airline business models and finally the relationship between customer satisfaction and on-time performance in the US aviation industry. To be able to answer questions we needed additional datasets to supplement our flight data. Analyzing the relationship between uncontrollable events and delays, we gathered hourly and daily weather condition data using the geocoordinates of the airports selected to explore. Regarding customer satisfaction, we scraped passenger review data across the airports and airlines in our dataset.

In the project planning phase, we hypothesized that uncontrollable events would be a major factor in explaining airline delays and cancellations. Our reasoning was due to the advancement of logistics management, and analytics will allow airlines to be run at a much more efficient scale. Our findings turned out to be the opposite. Our analysis concluded that there is a weak statistical relationship between moderate weather conditions and time delay. Indicating that operational inefficiency has a substantial impact on our target variables. However, we found that snowfall had a slightly more significant relationship when it comes to flight cancellations. Despite that, we did observe clear trends of weather-induced delays/cancellations during specific months of the year in a subset of the airports in the dataset. Our insights showed that weather conditions were not able to explain the variability in delay time and cancellations to a reasonable degree all the time, which implies that there are other factors impacting our target variables such as inefficient operations, and additional uncontrollable events such as system infrastructure failures, staffing shortages and security issues. In terms of customer satisfaction, we still believe that there is an explainable relationship concerning airline on-time performance. Unfortunately, the review data we came across did not offer any variance with regard to the sentiment observed. It is highly unlikely that customers will not respond to an improvement in airline performance and reputation. Notwithstanding, our analysis uncovered winners and losers in the US aviation industry, and the days and time you are likely to travel without much delay.

2. Dataset and Guiding Questions

2.1. Dataset

Data collected comes from three different sources, United States Department of Transportation, Skytrax-airlinequality.com, and Open-Meteo.com. USDOT meets the requirements of the Open Data Policy¹ while the Skytrax data forbids commercial usage without prior notice². Open-meteo uses an attribution 4.0 Creative Commons license which permits the ability to “share: copy and redistribute the material in any medium or format and adapt: remix, transform, and build upon the material.” United States Department of Transportation (USDOT) provides downloadable CSV files and webpages containing tables with up-to-date information about newly registered airports and airlines. Our core data is collected from “[Reporting Carrier On-Time Performance](#)”³. It is a table containing 10 years of flight data, from 2014-07 to 2024-06, with fields such as airline carrier, origin, destination, and whether the flight experienced any form of delay or cancellation. The dataset has 13 million rows, 24 columns, and a size of 2.86GB.

We have also collected three supplementary datasets from USDOT^{4 5 6}, containing the most up-to-date mapping between carrier code and the name of the carrier, airport code and the name of the airport, as well as tail number and the respective plane model. However, the core data contains mainly IDs for airports and carriers, which contain extremely high cardinality (24 airports and 18 airlines). To facilitate better visualization of our data and explore the possibility of utilizing Natural Language Processing, we have performed web scraping on one of the most influential and long-standing airport review sites, Skytrax, to acquire user comments and scoring related to different aspects of services of airlines and airports⁷. The web scraper was written using BeautifulSoup while the HTML search terms were created in reference to “[Skytrax-Scraper](#)”⁸ on Kaggle.

We did run into a unique problem of naming conventions during this part. Skytrax used informal and varied naming systems for different airports, for example:

Skytrax Name	USDOT Name
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¹ Data inventory. U.S. Department of Transportation. (n.d.). <https://www.transportation.gov/data>

² Site usage and legal information. SKYTRAX. (2024, January 4). <https://www.airlinequality.com/info/legal-use/>

³ Download Page - transtats.bts.gov. (n.d.-a).

https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr

⁴ World Airport Codes. World Airport Codes | Bureau of Transportation Statistics. (n.d.).

<https://www.bts.gov/topics/airlines-and-airports/world-airport-codes>

⁵ Airline codes. Airline Codes | Bureau of Transportation Statistics. (n.d.). <https://www.bts.gov/topics/airlines-and-airports/airline-codes>

⁶ Releasable Aircraft Database Download | Bureau of Transportation Statistics.(n.d.).

https://www.faa.gov/licenses_certificates/aircraft_certification/aircraft_registry/releasable_aircraft_download

⁷ Airline Reviews and rating. SKYTRAX. (2024a, June 26). <https://www.airlinequality.com/>

⁸ Notcostheta. (2024, January 3). Skytrax-scraper. Kaggle. <https://www.kaggle.com/code/notcostheta/skytrax-scraper>

Las Vegas Airport	Harry Reid International Airport (Formerly McCarran International Airport)
Los Angeles LAX Airport	Los Angeles International Airport

The unstable nature of this airport naming makes it incredibly difficult to match all our airport names to the correct review pages using simple string-matching algorithms. To tackle this problem, we have submitted names from Skytrax to Large Language Model (LLM) and requested a translation of formal airport names [[Appendix 1](#)]. The result had been verified by humans

to be relatively precise, with a few minor errors. Finally, open-meteo facilitates the ability to receive selected weather variables on an hourly, and daily basis given selected latitude and longitude pairs. We collected our airport geocoordinate pairs to get the most accurate weather conditions possible. The website offers an interactive skeleton API code that is adjusted based on the inputs selected. We chose the most relevant weather variables which included snowfall, rainfall, wind speed, wind gusts, wind direction, and the universal weather code describing the weather in each timeframe.

Our dataset contains 24 unique airports and their corresponding review sites. There are 18 airlines recorded by USDOT, and only 13 of them have dedicated reviews since airlines, such as PSA, operate as subsidiaries of other prominent airlines. This is a problem that our team needs to be mindful of during data analysis.

2.2. Data Cleaning

The basic procedures of data cleaning are documented below:

1. Perform basic data inspection using [ydata-profiling](#).
2. Data Cleaning ([HTML jupyter notebook](#)):
 - a. Removing duplicated columns
 - b. Converting to timestamp objects
 - c. Limiting analysis duration to 10 years
 - d. Replacing ID columns with descriptions
 - e. Creating column description meta data
3. Scrape and clean review data ([HTML jupyter notebook](#)).

Our code has been pushed to the following [GitHub Repository](#), for a better viewing experience, a [GitHub-Page Website](#) was also created. You may find our code and visuals used in the following reports in interactive form on the website.

2.3. Guiding Questions

In this section, we present our motivating guiding questions formulated in the project planning phase. These questions each pinpointed an area of US aviation and related on-time

performance that we aimed to analyze and provide insights on. We later discuss our findings in each of those guiding questions and how we pivoted when we ran into obstacles.

1. Do airlines with a hub and spoke business model have higher flight delays and cancellations than those with a point-point model such as low-cost carriers?
 - Unlike the point-point model, Hub operations are usually centralized serving many feeder and connecting flights. Such large-scale operations require tight scheduling and management of different operational activities, which are very complex.
2. What are the most recurring reasons for flight delays and cancellations?
 - We aim to discover the most frequent causes of flight delays and provide an analysis by airline, airport, and routes.
3. Is there a relationship between flight delays/cancellations and passenger satisfaction?
 - The key idea is to determine the correlation between flight delays/cancellations and passenger experience captured from online reviews. We will attempt to establish whether airline operators with lower flight delays enjoy higher customer satisfaction levels and vice versa.
4. How do US airlines and airports rank in on-time performance amongst each other?
 - We will drill down to compare the worst-performing airlines/airports in relation to their counterparts. What are the drivers of poor/good on-time performance? And how can stakeholders plan accordingly.

2.4. Division of Labour

Topic	Team Member
Weather	Youssef Abdelwahab
Airport + Aircraft	Geoffrey Wong
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Airline	Anthony Obi
Miscellaneous	Chibuzo Oha

3. Analysis

3.1. Airline Performance Review

The airport performance assessment was done across three key areas: passenger satisfaction rating as captured from the Skytrax online review, the arrival delay, and flight cancellations. Our review involved 12 US airlines comprising large commercial and small budget airlines.

3.1.1. Airline Passengers Review Ranking

Individual Scoring Attributes of Airline

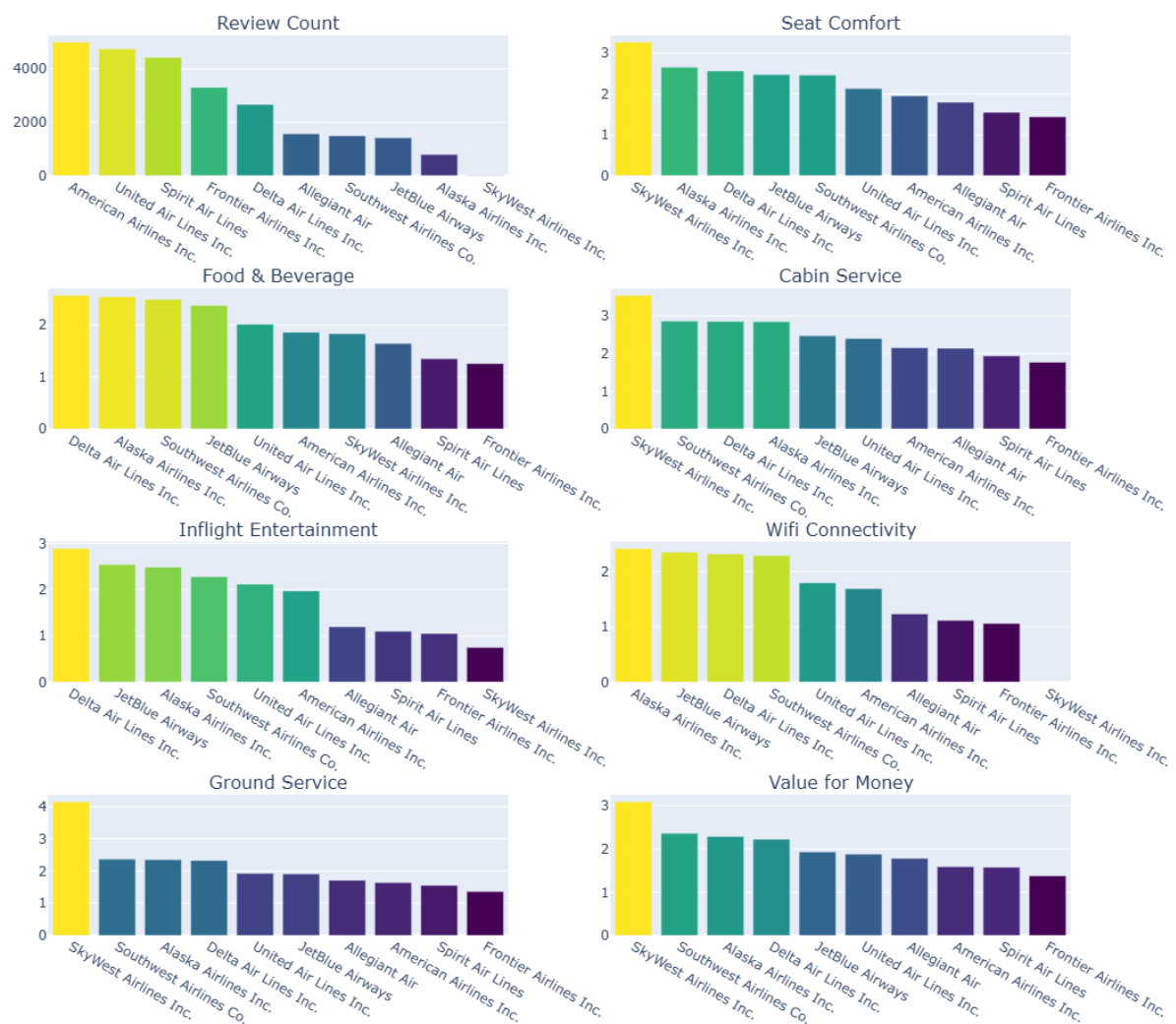


Fig. 3.1.1. Scoring of different categories of Airlines

As shown in the above figure, we have a total of 10 Airlines in our dataset with all the airlines receiving at least 5,000 reviews except Skywest airline which had only 41 reviews. The passenger ratings focused on 7 criteria, ground services, value for money, and seat comfort. Skywest outperformed the other airlines in 4 out of the 7 criteria, followed by Delta. However, given the limited number of ratings for Skywest (41 reviews) which is prone to small sample bias, we consider Delta as the overall winner in terms of passenger satisfaction rating.

3.1.2. Airline Arrival Delays

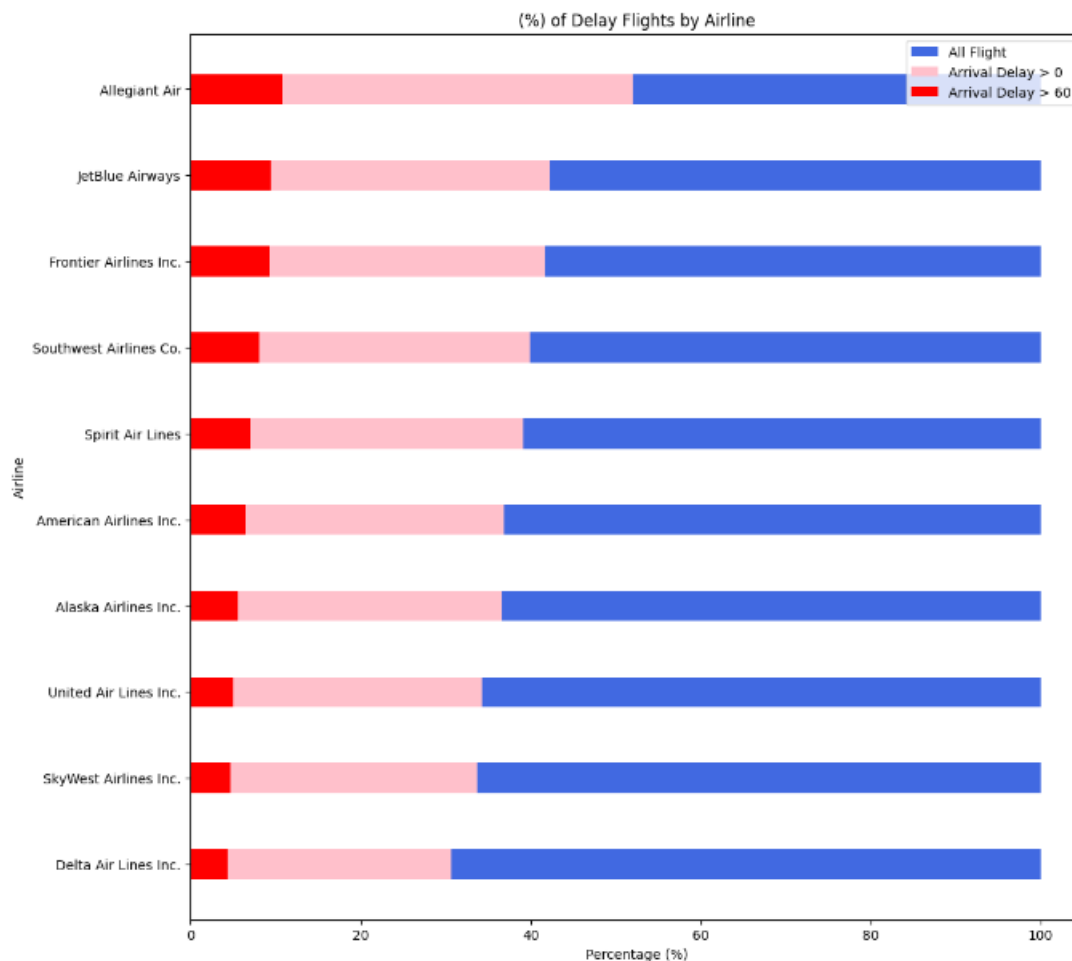


Fig. 3.1.2.1. Percentage of Delayed Flights by Airlines

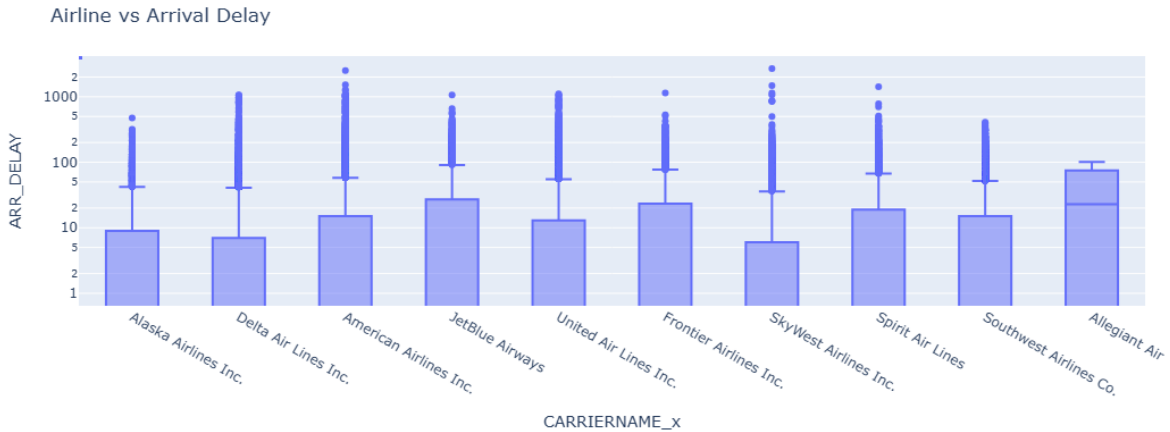


Fig. 3.1.2.2. Arrival Delay Distribution by Airlines

As seen from the two graphs above, Allegiant Air, a budget airline, has the worst arrival delays while Delta has the best airline on-time flight performance (lowest % flight arrival delays). Our analysis suggests that on average, around 50% of flights operated by Allegiant Air are delayed compared to only around 30% on Delta Airlines (Fig. 3.1.2.1). Furthermore, even though most airlines managed to maintain a median arrival delay of 0 minutes, Allegiant Air reported a dismal median delay of 20 minutes (Fig. 3.1.2.2).

We also noticed that most airlines have outliers with delays of up to 1,000 minutes, except for Allegiant. We hypothesize this could be due to cancellations.

3.1.3. Flight Cancellation

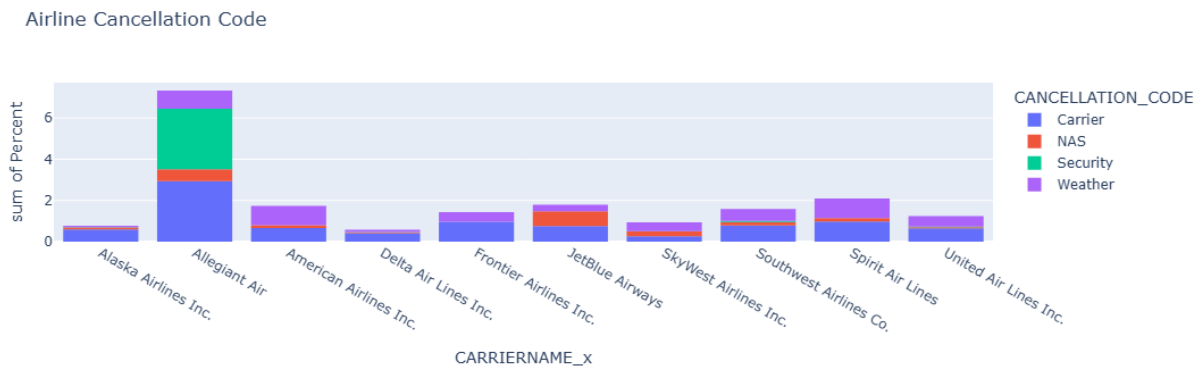


Fig. 3.1.3. Flight Cancellation by Airlines

Similar to the flight delay review, Allegiant Air has the worst performance in terms of flight cancellations. On average, 7% of Allegiant flights are canceled and Allegiant attributed a much larger portion of the cancellation to Security and Carrier (Fig. 3.1.3). The carrier delays match our hypothesis above. Instead of reporting severely delayed flights as carrier delays, Allegiant seemingly cancels them directly. However, the security cancellation is much more concerning. Although we have no data to explain what they are, the Federal Aviation Administration⁹ states these delays are likely due to Security breaches or even evacuation of terminals or concourses. We would like to investigate it in the future, but the situation sounds dangerous regardless.

Likewise, Delta performed very well on this important flight metric with fewer flight cancellations compared to the other airlines.

3.1.4. Conclusion for Airline

Overall, our analysis indicates that Delta Airlines is the best airline based on passenger ratings, on-time performance, and flight cancellations, delivering superior passenger experience and better on-time flight performance.

3.2. Airport On-Time Performance

3.2.1. Airport Review Ranking

⁹ *Types of Delay.* Federal Aviation Administration.
https://aspm.faa.gov/aspmhelp/index/Types_of_Delay.html

Individual Scoring Attributes of Airports



Fig. Airport Review Scores Ranking

On Skytrax, the commenters are required to fill in an overall score, whether they would recommend the airport, and 8 additional scores for aspects of the airport, including queue time, terminal sign, and airport staff, of which we hypothesized to have significant impacts on the delay and cancellations related to said airport.

From the score, we discovered that, in general, medium-sized airports are favored over large hub airports. Amongst them, Portland (PDX), San Jose (SJC), and John Wayne (SNA) took the first place in review scores in different categories. While world-famous airports, like the Los Angeles Airport (LAX), received negative ratings across almost all categories

3.2.2. Origin Airport Arrival Delay

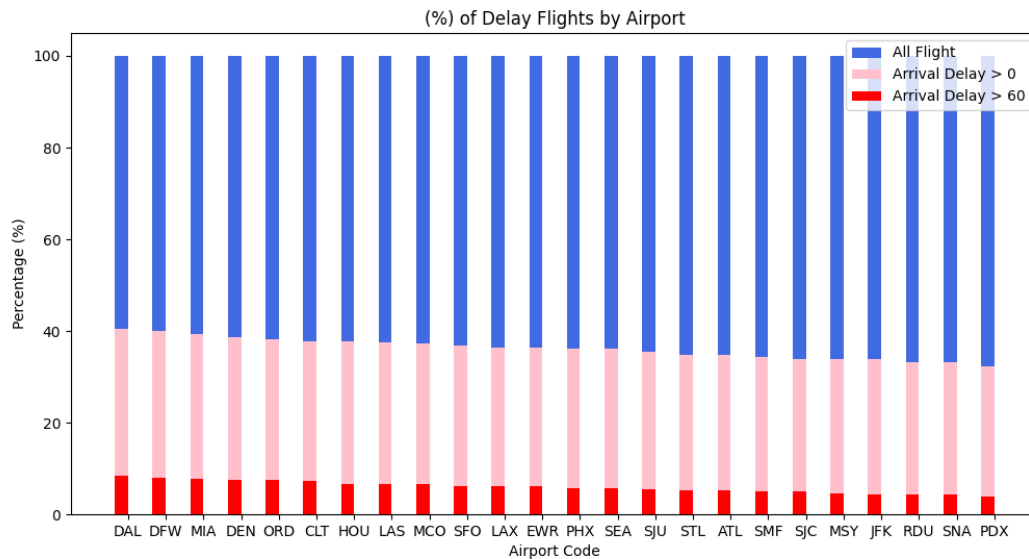


Fig. 3.2.2.1. The chart shows the percentage of slight delay and severe (>1 hour) delay in different origin airport

By comparing airport reviews with the percentage of delayed flights from each of the original airports, we feel comfortable with suggesting PDX, SNA, and SJC airports due to their stellar performance in both user sentiment and on-time performance (Fig. 3.2.2.1).

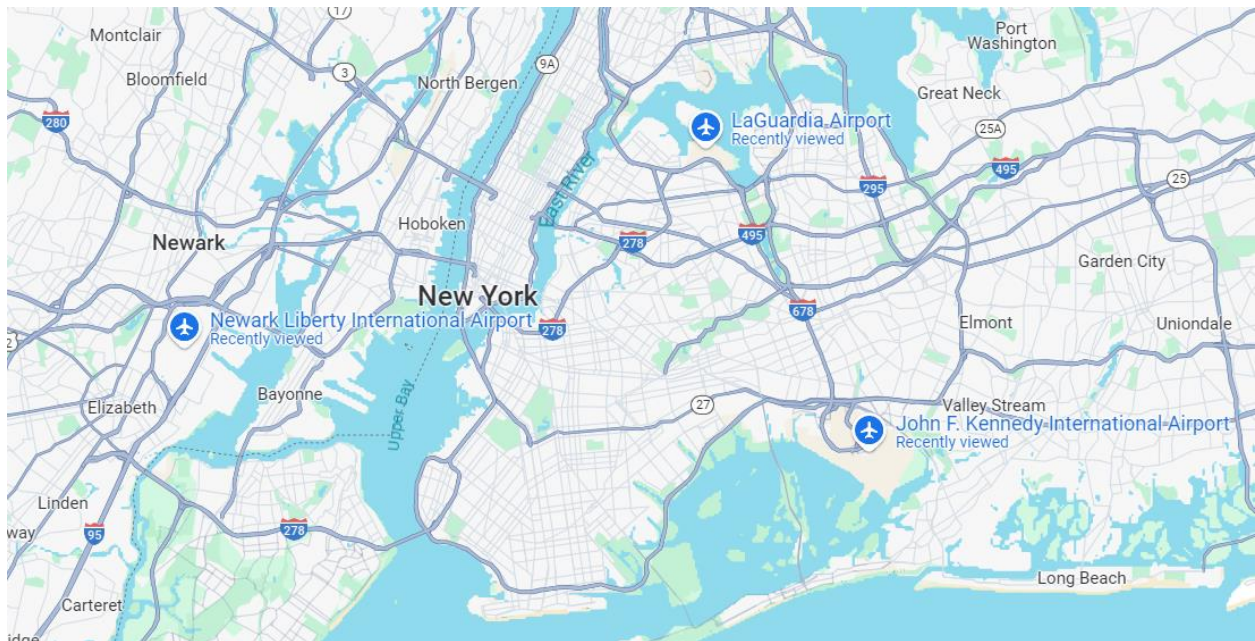


Fig. 3.2.2.2. Map of New York City

On top of that, we also discovered that among the airports in New York City, JFK, despite being busier, has fewer delayed flights than Newark Liberty International Airport (EWR).

Considering these 2 airports are only 1 hour away from one another (Fig. 3.2.2.2), it is safe to recommend JFK over EWR whenever possible.

3.2.3. Airport Review Correlation with Delays

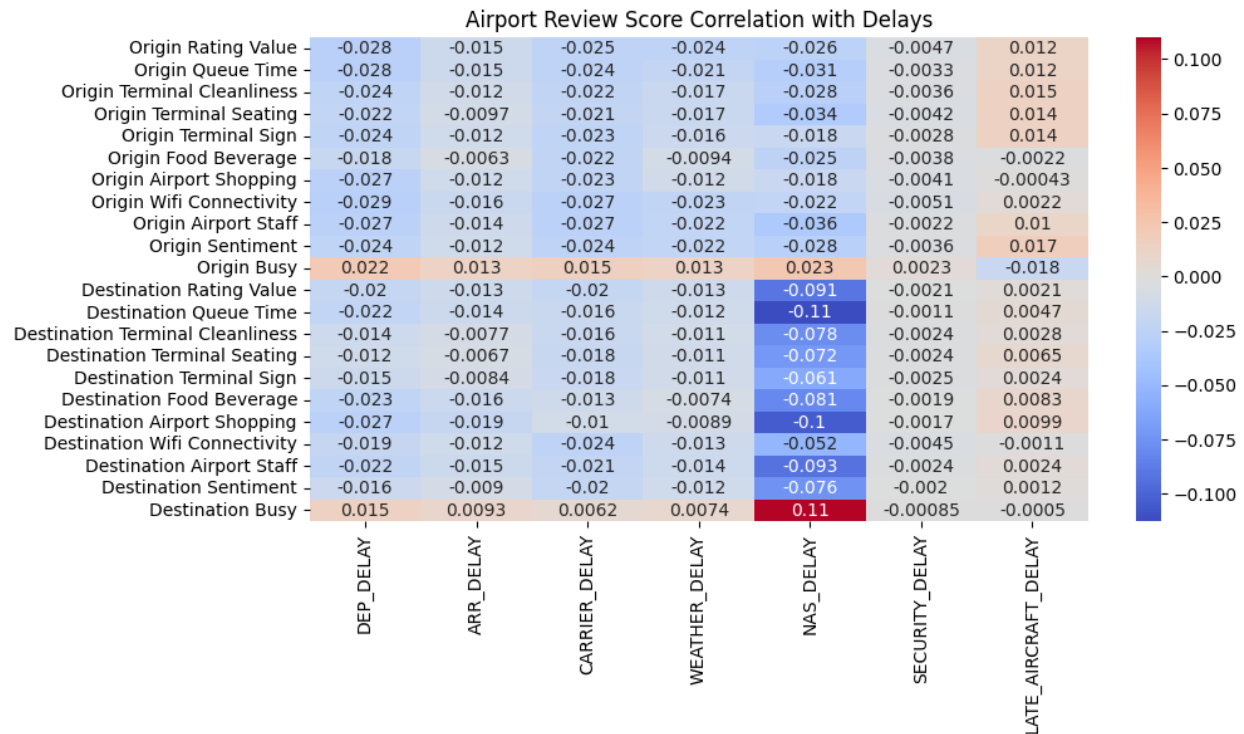


Fig. 3.2.3. Airport Review Correlation with Delays

Our dataset contains 7 different values for delays. The 2 major ones:

Departure Delay	The flight left later than expected.
Arrival Delay	The flight arrived later than expected.

And 5 subcategories explaining why the flight was delayed:

Carrier Delay	Controlled by the airlines, such as aircraft cleaning and maintenance.
Weather Delay	Extreme weather conditions
National Airspace System (NAS) Delay	Delay controlled by the NAS, such as nonextreme weather, heavy traffic volume, or air space control.
Security Delay	Security breach, screening equipment malfunction, or even evacuation of terminal or concourse.
Late Aircraft Delay	The ripple effect from previous late arrival of the same aircraft.

In the graph above, we discovered that among the 5 subcategories of delay, NAS delay has the strongest correlation with review scores of the destination airport in general. More specifically, the higher the review scores and the less busy the destination airport is, the lower the NAS delay. Although the correlation is only around 11%, this matches the description of NAS delay.

We also noticed there is a very consistent pattern between the busier airport and reviews. Specifically, if the airport is a busy, hub airport, such as LAX, typically the review score is relatively low.

3.2.4. Correlation Among Reviews

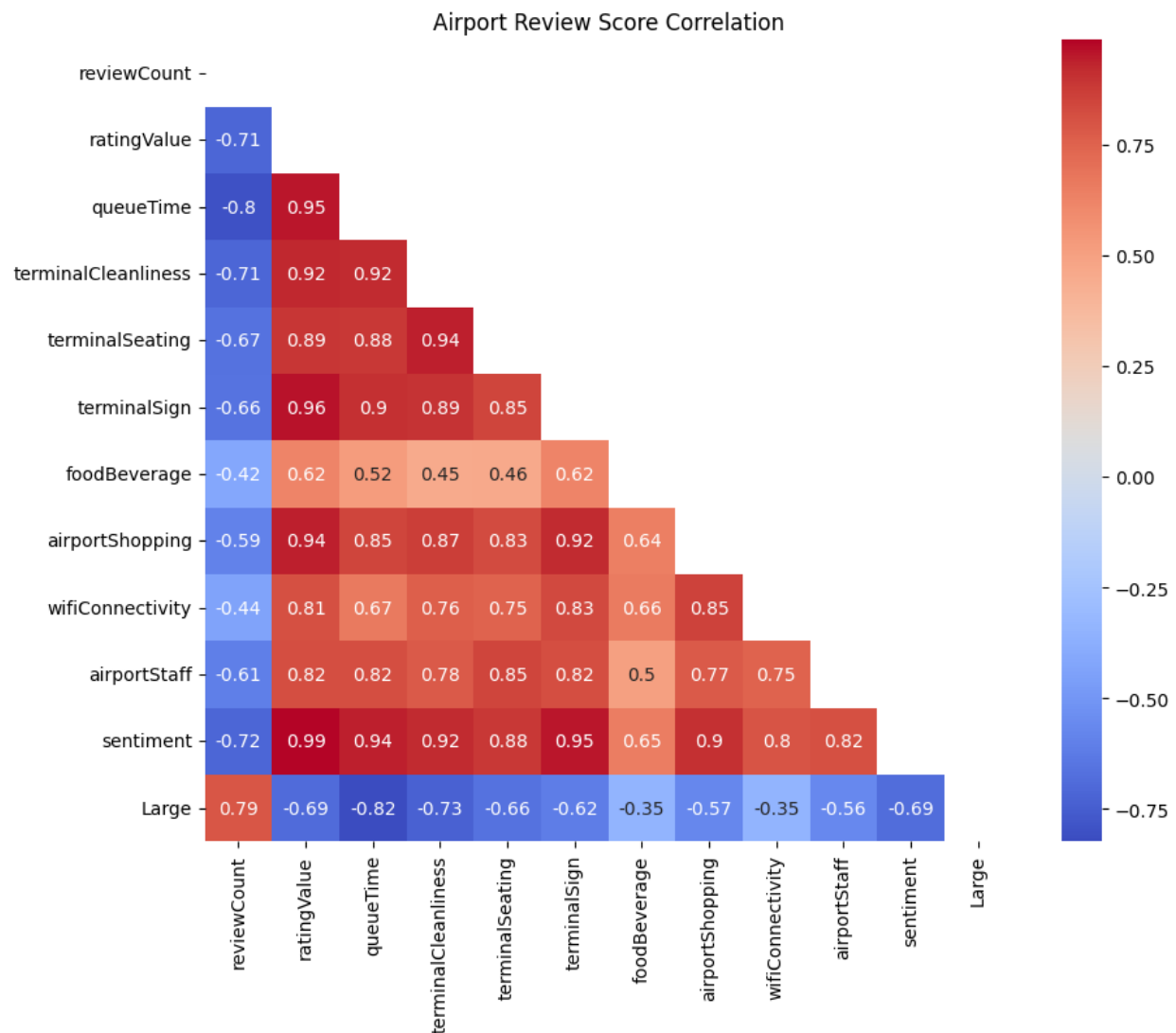


Fig. 3.2.4. Correlation among Airport Reviews

Examining the patterns found in the reviews further, we have uncovered a significant pattern. Despite there being 8 different subcategories of reviews, they have strong correlations, sometimes up to 99%, between one another. This is a very typical survivor bias found amongst reviews since only the people with the strongest emotion, usually negative, would take their time leaving reviews. Additionally, reviewers holding overall negative sentiments would usually have bad impressions of all aspects of service.

In conclusion, despite collecting review scores about different aspects of airports, the review bias shrank our data variability considerably. The review data only seems to show a 1-dimensional view: How busy the airport is, which negatively correlates with the review score.

3.2.5. Destination Airport NAS Delay

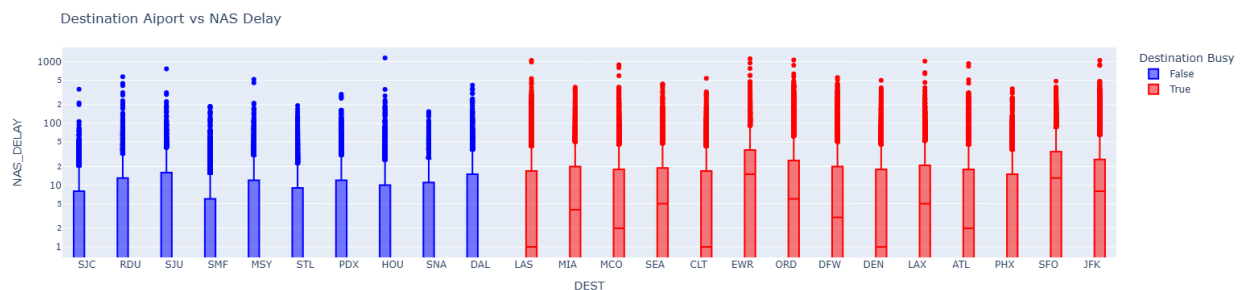


Fig. 3.2.5. Destination Airport NAS Delay

Despite the data quality of reviews from Skytrax, the correlation scores do point us towards the direction of investigating the NAS Delay with respect to the destination airport. The chart above shows the distribution of NAS delay in different airports.

We can see that the median NAS delays of small airports (blue columns) are all 0 minutes, while the big hubs have a median NAS delay of around 2 to 10 minutes. This led us to conclude that, whenever possible, it is a good idea to avoid flying to larger hub airports if you would like to have a higher chance of arriving on time.

3.3. Routes

3.3.1. Hypothesis on Route Traffic and Delays

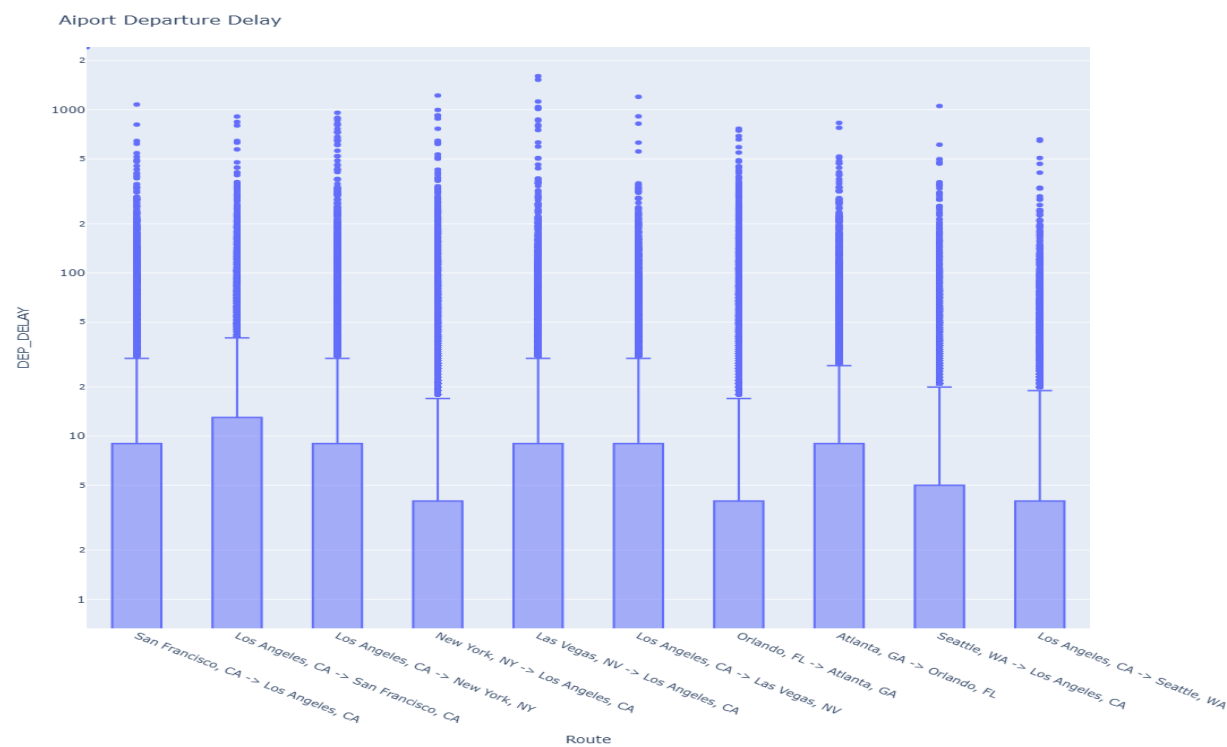


Fig. 3.3.1.1. Top 10 Busiest Routes Departure Delays

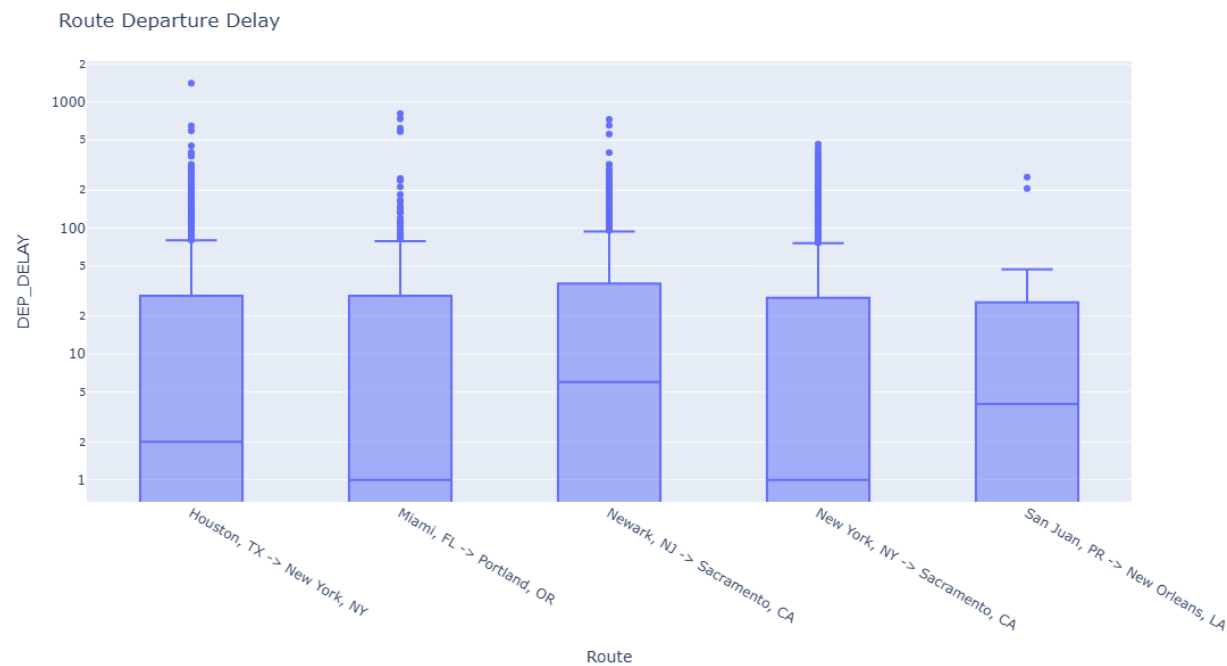


Fig. 3.3.1.2. Top 5 Most Delayed Routes

We hypothesized that **"the busier the route, the more delays it will experience."** To test this, we examined the top 10 busiest routes, and found, surprisingly, all of them have a median departure delay of 0 minutes (Fig. 3.3.1.1). Contrary to our initial belief, the data revealed that most delays occurred on less busy routes. For instance, the Houston to New York route, which only had one flight per day, experienced significant delays (Fig. 3.3.1.2), disproving our initial hypothesis.

3.3.2. Route Use Frequency vs Median Delay

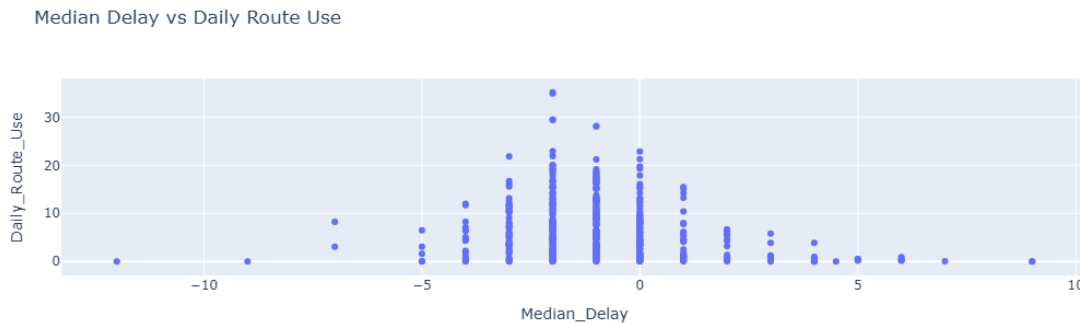


Fig. 3.3.2. Daily Route Usage vs Median Departure Delay

From the most delayed routes above, we have found our initial hypothesis of crowded routes being more delayed to be completely wrong. By looking into the pattern of delays more closely, we found that the delay of busy routes concentrates around -2 minutes delay while flights that involve medium airports have greater error range, spreading from -10 to +10 minutes.

This deviation is likely due to infrequent routes getting both less data and attention for readjustment when it comes to expected flight time.

3.4. Weather Impact

One of our guiding questions was to explore the impact of external uncontrollable events on the on-time performance of US aviation. The United States is known to experience extreme weather systems that develop from the southern hemispheres. We hypothesized that uncontrollable events such as adverse weather will explain a large portion of poor on-time performance.

Our dataset included different types of delay columns, 'Carrier Delay', 'Weather Delay', 'NAS Delay', 'Security Delay' and 'Late Aircraft Delay' each indicating the proportion of the total delay time the flight encountered. We opted to combine them all into one column as 'Total Delay' as we believe that the more passengers are stuck in the airport, the higher the load on airport facilities and infrastructure thus creating a bottleneck across all aspects of flying. Additionally, we chose to round the 'Arrival and Departure Times' of the flights to the nearest hour as we merged them to hourly airport weather conditions. We chose to do this to capture the most accurate weather

conditions at the time of the flight. Once that was complete, we grouped our dataset by Flight Date, Flight Hour, and Airport Code.

3.4.1. Impact on Delays

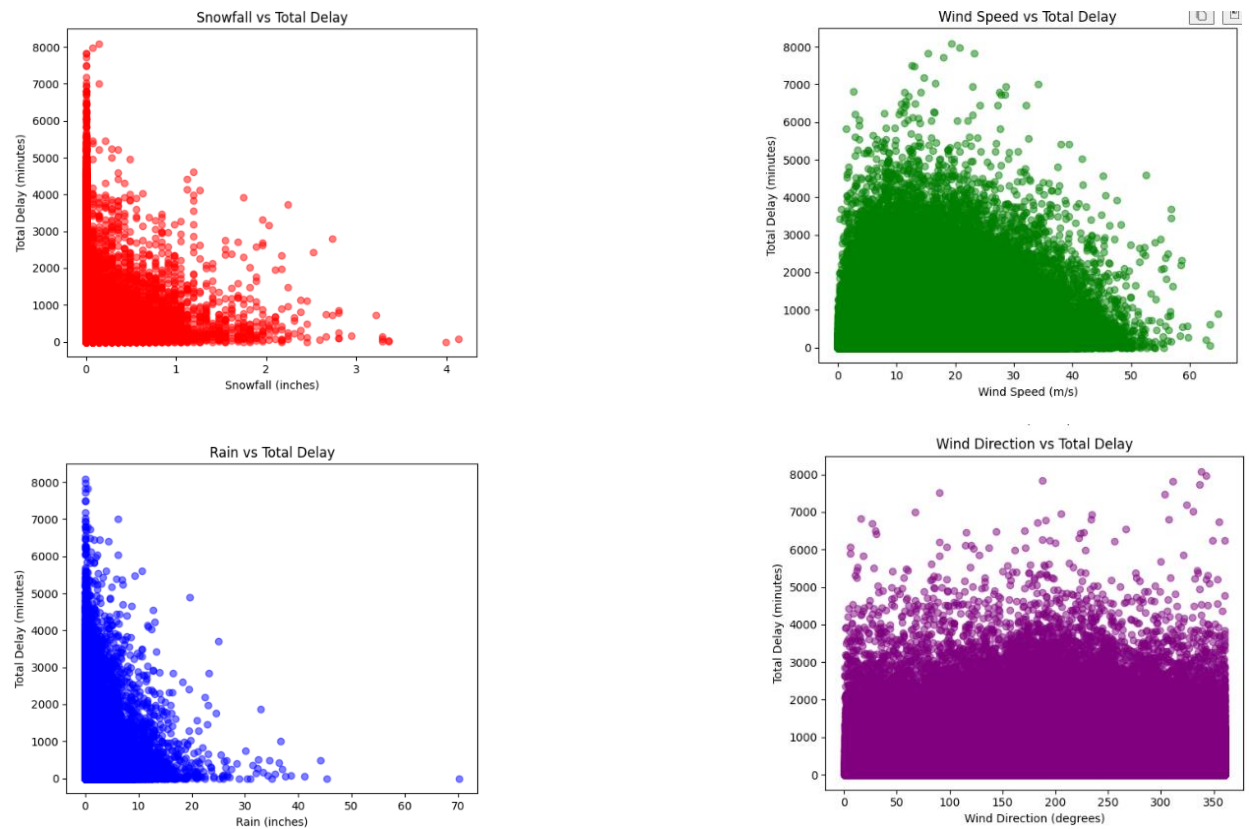


Fig. 3.4.1.1 Scatterplot of weather variables with delay times

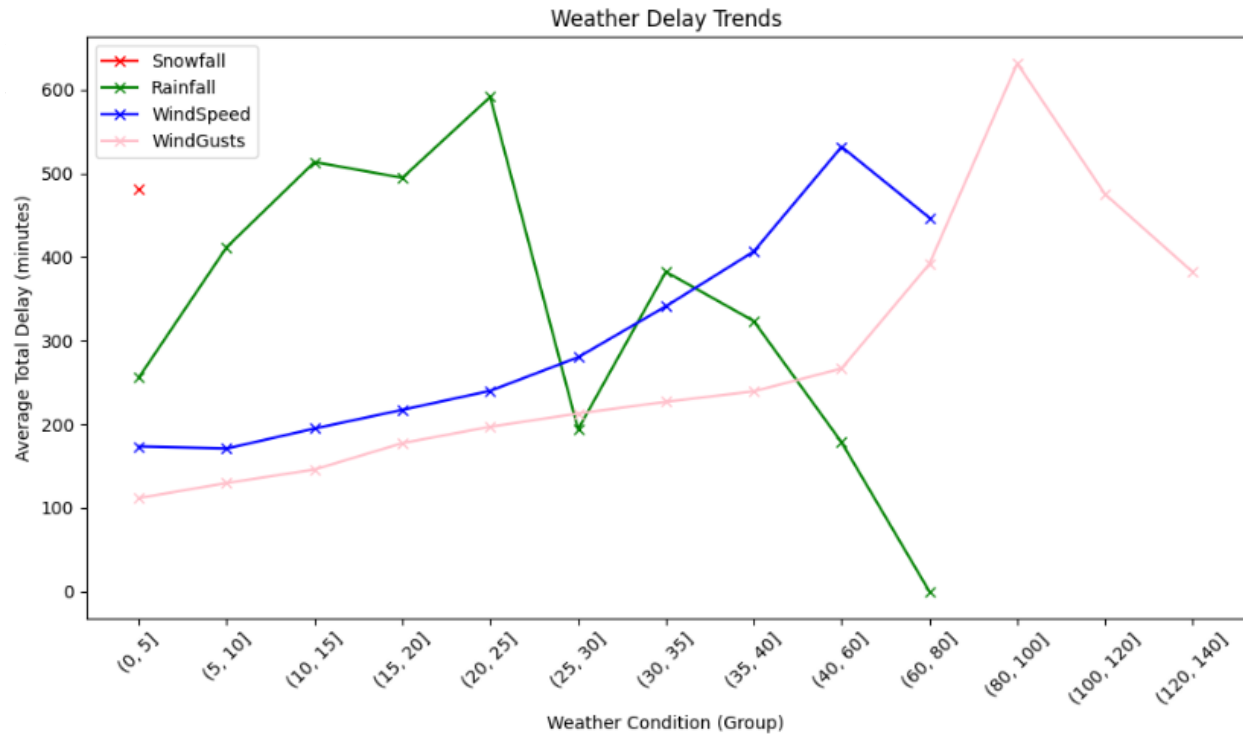


Fig. 3.4.1.2 Trend Analysis of Ranges of Weather Variables and Time Delay

In this visual, we aggregated the mean delay times across all ranges of the observed weather conditions in our dataset. We see a clearer picture when we look at the trend analysis. Looking at the red point on our plot we see that snowfall does have an influence on increasing delay times, however, we observe only one data point. This means that snowfall is observed in certain airports only and that we might have much fewer flights happening on those days thus more outright cancellations which reduces our observed data points. We also see a consistent trend with wind speeds and wind gusts. Delays increase as the speed of wind increases; however, we see a sharp drop in delay times at high wind speed ranges. This again could indicate that fewer flights occur at those ranges. Finally, looking at rainfall we see an interesting pattern. We initially see higher delay times as rainfall increases but we then observe a sharp drop in delay times at 25-30mm of rain, once again indicating fewer flights. Nonetheless, we see a sharp rise in delay times at an even higher rainfall rate. This insight might explain the different operational capabilities in different airports, as some can handle harsher weather conditions than others.

3.4.2. Impact on Flight Cancellations

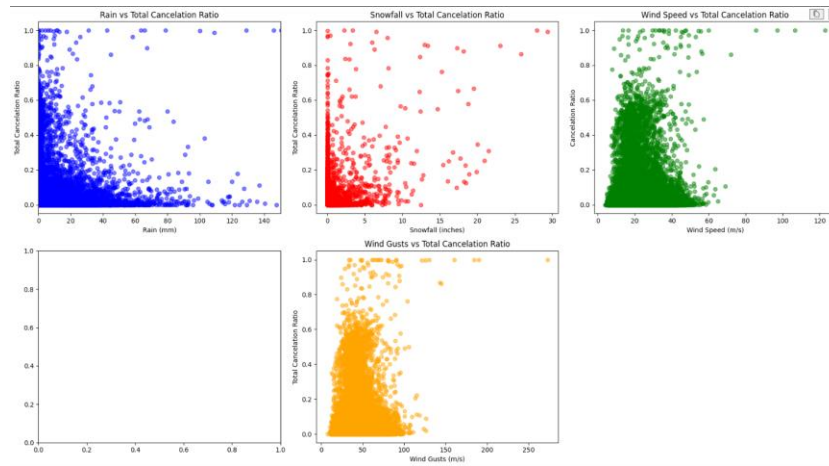


Fig. 3.4.2.1. Scatter plot of Weather Variables and Flight Cancellation Ratio

we see that the sum of snowfall on a given day as the strongest relationship between all weather conditions followed by the sum of rain fall and then finally the max wind speed and gusts. These relationships can be visualized through our scatter plot graphs (Fig. 3.4.2.1).

Given the positive relationship between the flight cancellation ratio and sum of snowfall, we wanted to explore how each airport fares with snowfall events to try to provide some recommendations. We did this by graphing the average cancellation ratio of all airports across the ranges of observed total snowfall.

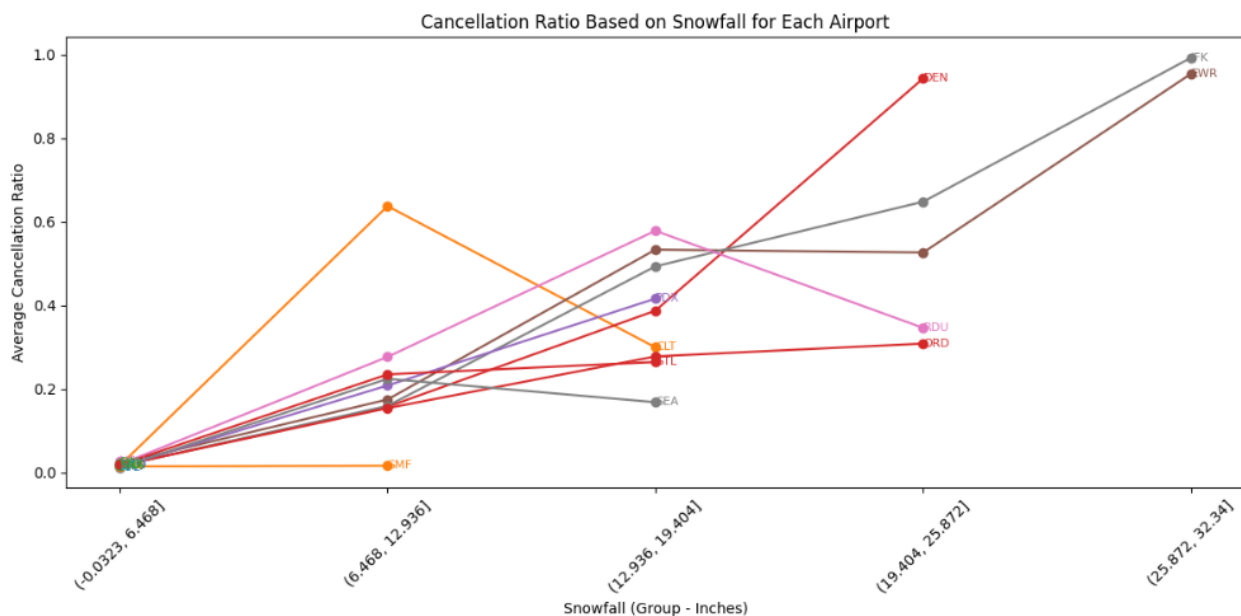


Fig. 3.4.2.2. Flight Cancellation Ratio Trend Across Observed Airports

As we can see, not all airports in our datasets appear on this plot. Certain regions in the United States experience snowfall at varying degrees than others within the months of winter. Most notable ones being New York’s John F Kenedy and Newark, Denver, Chicago International Airports.

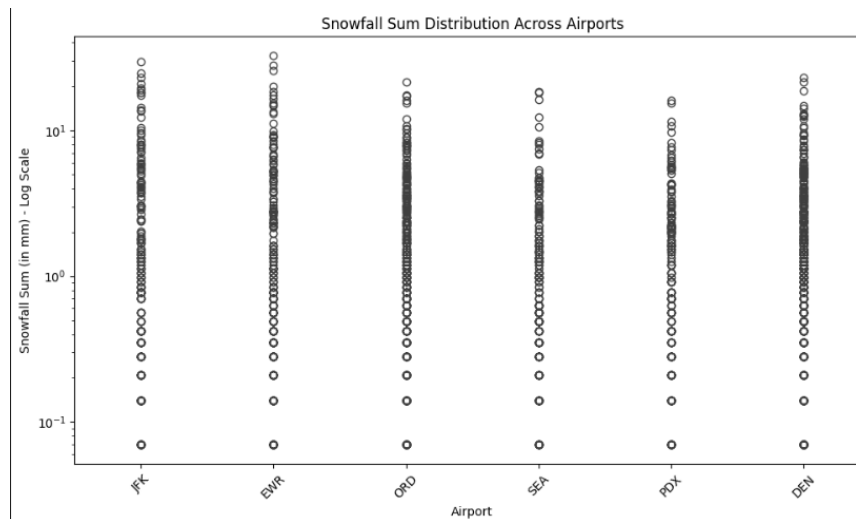


Fig. 3.4.2.3 Distribution of Sum of Snowfall Across Selected Airports

4. Discussion & Conclusion

4.1. Guiding Questions and Findings

<i>Guiding Questions</i>	<i>Findings</i>
Would routes that are busier be more prompted to delays?	No. Routes that are busy have more data to optimize their flight duration estimation, while routes that are sparsely flown have a larger standard deviation of delay (Sec. 3).
The most frequent causes of flight delays.	The correlation is generally very low (~10%). However, we observed a few standouts: - <i>Allegiant</i> is disproportionately bad, while the rest of the airlines are relatively similar - Hub Airport Destination has high NAS Delay (Sec. 3.2.3). -
Is there a relationship between flight delays/cancellations and passenger satisfaction?	Slightly. Well reviewed medium airports, like PDX, SJC, and SNA also have good on-time performance (Sec. 3.2.2).
How do US airlines and airports rank in on-time performance amongst each other?	Whenever possible and prices permit, choose Delta Airline, and fly to a medium-sized airport (Sec. 3.2.4) for better on-time performance.
What are the most recurring reasons for flight delays and cancellations?	Weather conditions are not the most frequent cause of delays/cancellations. However, the winter months in JFK, EWR, ORD, and DEN will experience significant

disruptions on days with moderate to heavy snowfall.
([Sec 3.4.2](#))

4.2. Other Findings

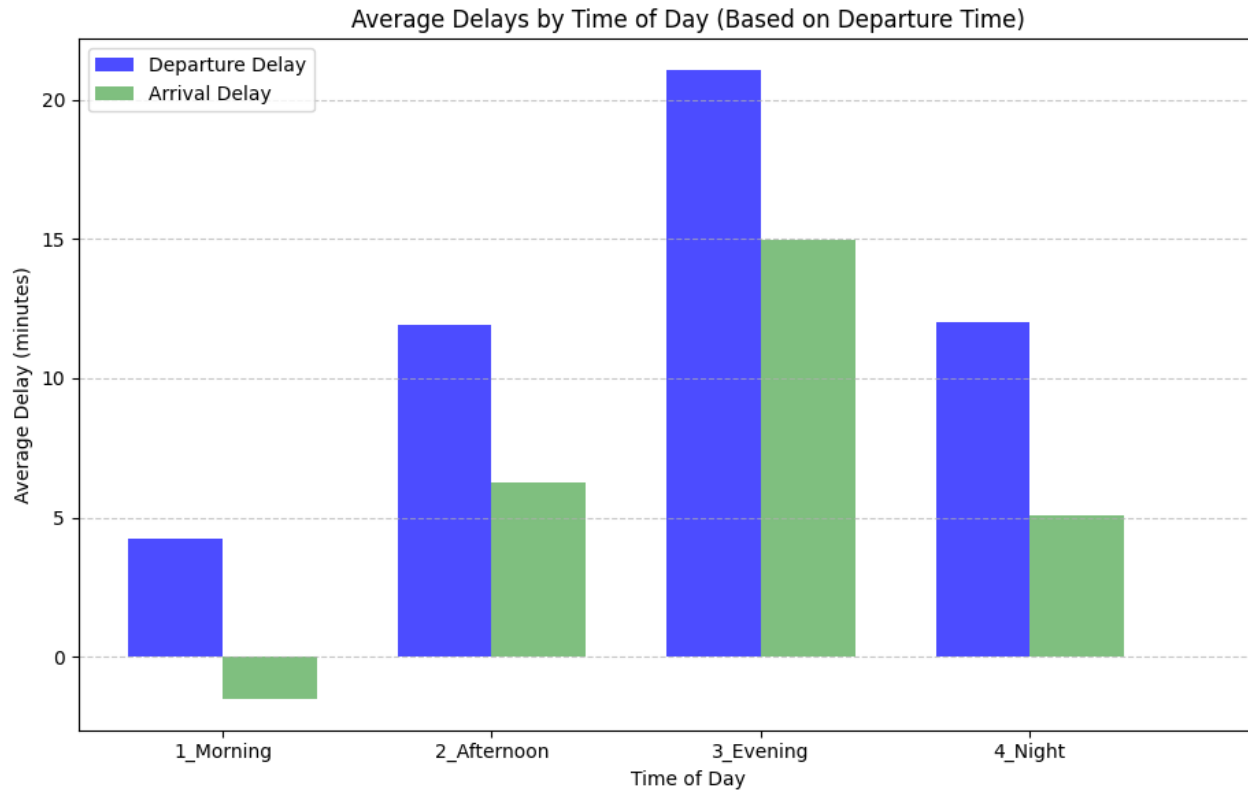


Fig. 4.2. Average Delay by Time of Day

Flights in the **evening (6 PM to midnight)** experienced significantly higher delays compared to morning flights. One could assume that this is likely due to congestion and workforce shortages. The recommendation is to address operational challenges for evening flights, including potential staffing improvements.

4.3. Obstacles

Review Data Correlation in Airport (Sec. 3.2.4)

Description: We scraped review scores to give our data more dimensions, ideally capturing a specific aspect leading to delays and cancellations. However, we found that the scores show a singular pattern. **The busier the airport is, the more negative the rating it receives.** Ultimately, we are only able to find the relationship between business and NAS delay.

Future Actions: We hope to acquire officially recorded, objective statistics about each airport, similar to the aircraft specification.

Lack of Data for Unanticipated Events

Description: The data we collected ultimately have a relatively low correlation with delays and cancellations. This leads us to hypothesize that the major bottlenecks of flights are not consistent factors, such as staff quality and general climate, but sudden events, such as system failures and airport incidents.

Future Actions: We hope to acquire time-sensitive events related to airports, including employee turnover overtime, labour strikes, and sudden changes in weather conditions.

4.4. Conclusion

The data exploration provided valuable insights into various aspects of aviation performance, particularly highlighting the role of weather, airline practices, and airport size in influencing delays. While the original hypothesis regarding Hub-and-Spoke vs. Point-to-Point delays remains inconclusive, clear findings emerged in other areas, such as the negative impact of weather conditions on delays and the superior performance of certain airlines like Delta. Additionally, the busiest routes did not show significant delays, challenging the assumption that busier routes inherently face more disruptions.

4.5. Recommendation

- **Weather Preparedness:** Airports and airlines operating in regions prone to snowfall and high winds should enhance their weather contingency plans to mitigate delays and cancellations. Air passengers should also expect increased chances of flight disruptions in months with frequent snowfall.
- **Evening Flight Management:** Airlines should consider improving workforce planning and operational strategies for evening flights, which tend to suffer from higher delays.
- **Customer Experience vs. Performance:** We do believe that in a highly competitive market, consumers will shift their choices to businesses that will provide them with value and effectively meet their needs. In the scope of passenger reviews, our dataset observed a significant positive skew, hence most of the reviews were negative despite certain airlines clearly improving their performance. Due to this we were unable to model the relationship between customer satisfaction and airline on-time performance.
- **Smaller Airport Optimization:** Given that smaller airports handle NAS delays better, larger airports could explore strategies from smaller airports to improve their own traffic management systems.

5. References

- [1] *Data inventory*. U.S. Department of Transportation. (n.d.).
<https://www.transportation.gov/data>
- [2] *Site usage and legal information*. SKYTRAX. (2024, January 4).
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6. Appendix

6.1. Prompts for Correcting Airport Names

Prompt
I have a list of airport names but they are not official, given the list of names I provided, return a csv file containing the unofficial name and the formal name, "NICKNAME", "ACTUAL_NAME" for each of the given airport: {Airport_Names}

